BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, HYDERABAD

INTERMEDIATE (11TH CLASS) EXAMINATION, 2023

PHYSICS

Past Paper 2023

Retyped faithfully from the original board paper — same questions, no changes.

Multiple Choice Questions (M.C.Qs)

Note: Q.No.1. Choose the correct answer for each from the given options.

Q1. Al-Shifa an encyclopedia of philosophy was written by ------

(a) Al-eruni
(b) Ibn-e-Sina
(c) Ibn-ul-Haitham
(d) Al-Razi

Q2. Pound is the unit of:

(a) Mass
(b) Energy
(c) Weight
(d) Torque

Q3. The dot product between two identical unit vectors is equal to -------.

(a) i
(b) Zero
(c) One
(d) 2i

Q4. The components of a vector are such as x is +ve and y is –e then vector lies in ------ quadrant.

(a) 1st
(b) 2nd
(c) 3rd
(d) 4th

Q5. The dimensions of angular velocity, frequency is:

(a) $M^0L^1T^1$
(b) $M^1L^1T^{-1}$
(c) $M^0L^0T^{-1}$
(d) $M^1L^1T^{-2}$

Q6. ------ of the following is one dimensional motion.

(a) The wheels of a moving train
(b) The earth revolving round the sun
(c) A train running on a straight track
(d) Electron moving around the nucleus

Q7. The speed of projectile v at any instant is equal to -------.

(a) $(v_x^2 + v_x^2)$
(b) $(v_x^2 + v_y^2)^2$
(c) $(v_x^2 + v_y^2)^{\wedge}1/2$
(d) $(v_x + v_y)^2$

Q8. It is assumed that in projectile motion the acceleration along vertical direction will ------

(a) Decrease
(b) Increase
(c) Remain the same
(d) None of these

Q9. The time period of rotating body is given by:

(a) $T = 2\pi\omega$
(b) $T = 4\pi^2\omega$
(c) $T = 4\pi^2\omega^2$
(d) $T = 2\pi/\omega$

Q10. A person whose weight is 150 pound on the earth, on the moon his weight will be approximately -------.

(a) 20 pounds
(b) 30 pounds
(c) 120 pounds
(d) 150 pounds

Q11. The term torque is synonymous with -------.

(a) Moment of force
(b) Moment of inertia
(c) Angular momentum
(d) Linear momentum

Q12. The moon complete one revolution around the earth in -------.

(a) 365.3 days
(b) 29.5 days
(c) 27.3 days
(d) 30 days

Q13. The KWh is the unit of -------.

(a) Work
(b) Power
(c) Pressure
(d) None of these

Q14. When the length of pendulum doubled, its time period becomes -------.

(a) Halved
(b) 1.41 times
(c) Doubled
(d) Remain same

Q15. A form of energy that produced sensation in our auditory system is known as -------.

(a) Phase
(b) Wave front
(c) Sound
(d) Sensational energy

Q16. E-rays are not diffracted though grating because X-rays having -------.

(a) Very short frequency
(b) Very short wavelength
(c) Large Amplitude
(d) None of these

Q17. The erect image by convex lens is possible when distance of object is ----.

(a) Less than focal length
(b) Equal to focal length
(c) Greater than focal length
(d) Equal to double the focal length

Q18. The author of Kitab-ul-Masoodi was -------.

(a) Al-Beruni
(b) Ibn-e-Sina
(c) Ibn-ul-Haitham
(d) Al-Razi

Q19. Light year is the unit of -------.

(a) Light
(b) Distance
(c) Time
(d) None of these

Q20. The division of a vector $\vec{A}$ by a positive number in is given by $\vec{B} = |m|\vec{A}$ where m = 1/n the direction of $\vec{B}$ is ------

(a) Same as $\vec{A}$
(b) Opposite to $\vec{A}$
(c) Perpendicular to itself
(d) None of these

Q21. If resultant of two forces having magnitudes 13N and 13N acting on a point is 13N, then angle between two forces is -----

(a) 0°
(b) 60°
(c) 90°
(d) 120°

Q22. If 1 kg and 1 gm Massive bodies are dropped from the same height simultaneously then ------- (Assume friction is zero).

(a) 1kg will reach the ground earlier
(b) 1gm will reach the ground earlier
(c) Both will reach the ground at the same time
(d) Depends upon nature of the mass

Q23. The property of matter by virtue of which it remains or tries to remain in its original state is known as -------.

(a) Force
(b) Inertia
(c) Friction
(d) Viscosity

Q24. The velocity of projectile is maximum at -------.

(a) Target
(b) Point of projection
(c) Both (a) & (b)
(d) Highest point

Q25. Centripetal force and centrifugal force are always -------.

(a) Parallel to each other
(b) Anti parallel to each other
(c) Perpendicular to each other
(d) None of these

Q26. The S.I unit of Angular momentum is -------.

(a) N.S
(b) J.S
(c) J.m
(d) N.m

Q27. The dimensions of gravitational constant are -------.

(a) $L^3M^{-1}T^{-2}$
(b) $L^2M^2T^{-1}$
(c) $L^1M^{-1}T^{-1}$
(d) $L^1M^{-2}T^{-3}$

Q28. If the velocity of body "A" is double than body "B" both having same mass then the ratio of K.E of both is --------.

(a) 1 : 2
(b) 2 : 21
(c) 4 : 21
(d) 1 : 4

Q29. A body executes S.H.M having maximum velocity -------.

(a) $v_0 = x\sqrt{(k/m)}$
(b) $v_0 = x_0\omega$
(c) $v_0 = x - v_x/\sqrt{(x = x^2/x_0^2)}$
(d) All of these

SECTION-B (SHORT QUESTIONS)

> Note: Answer any SEVEN of the following questions. Each question carries 05 marks.

Q2. Prove that the formula $s = 1/2\ at^2$ is dimensionally correct. Where s is distance, a is acceleration and t is time.

Q3. What is the relation between angular and linear quantities?

Q4. At what condition direction of torque be reverse?

Q5. Prove the law of conservation of momentum.

Q6. How the mass of earth determined?

Q7. What are the conditions for interference of light?

Q8. Why the amplitude of simple pendulum kept small?

Q9. Find the unit vector parallel to vector $\vec{A} = 4\hat{i} + 7\hat{j} + 4\hat{k}$.

Q10. A boy throws a ball upward with speed of 14.7 m/s. How long the ball rises?

Q11. An ambulance travels to observer with speed 80 km/h, its siren emits sound at frequency of 2000 Hz what frequency will be detected by stationary observer?

SECTION-C (DESCRIPTIVE QUESTIONS)

> Note: Answer any TWO of the following questions. Each question carries 09 (5+4) marks.

Q12. (a) State and explain product of two vectors. Show that dot product is commutative. (b) What is projectile motion? Drive the formula for its maximum range.

Q13. (a) What are characteristics of simple harmonic motion? Prove that the motion of mass-spring system is simple harmonic motion. (b) How the magnitude and direction of resultant can obtain by addition of vectors by rectangular component method.

Q14. Write notes on any TWO of the following: (a) Thin lens formula (b) Simple pendulum (c) Universal law of Gravitation (d) Michelson’s interferometer